[redacted]

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1. [redacted]

[redacted]

$$\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = 0 \tag{1}$$

[redacted] $\mathcal{L}$ [redacted](Lagrangean)[redacted] q_i [redacted] $\dot{q}_i$ [redacted](**[redacted]**) [redacted]

[redacted]

$$L = T - V \tag{2}$$

[redacted] T [redacted] V [redacted]

[redacted]

1.1. [redacted]

[redacted] x [redacted] [redacted] $x(t)$ [redacted] [redacted] $t = t_0$ [redacted] x_0[redacted] $t = t_1$ [redacted] x_1 [redacted] $t_0 < t < t_1$ [redacted]

[redacted] $f(x, \dot{x})$ [redacted] [redacted] $x(t_0) = x_0$ [redacted] $x(t_1) = x_1$ [redacted](Action)[redacted] [redacted]

$$A = \int_{t_0}^{t_1} f(x(t), \dot{x}(t))\,\mathrm{d}t \tag{3}$$

[redacted]

1.2. [redacted]

[redacted]

[redacted] x [redacted]

[redacted]

[redacted]

[redacted] $f(x, \dot{x})$ [redacted] A [redacted] [redacted] δ [redacted]

$$\delta A = \int_{t_0}^{t_1} \delta f(x, \dot{x})\, \mathrm{d}t = 0 \tag{4}$$

[redacted]

[redacted] $\delta f(x, \dot{x})$ [redacted] f [redacted] [redacted] x [redacted] x_i [redacted] [redacted] t [redacted] ε [redacted]

$$f\left(x_i, \frac{x_i - x_{i-1}}{\varepsilon}\right). \tag{5}$$

[redacted]

[redacted] i [redacted] x_i [redacted] f [redacted] [redacted] f [redacted] i [redacted] f_i [redacted] x_i [redacted]

$$\begin{aligned} \frac{\partial}{\partial x_i} &= \frac{\partial}{\partial x_i} f\left(x_i, \frac{x_i - x_{i-1}}{\varepsilon}\right) \\ &= \frac{\partial f_i}{\partial x_i} + \frac{\partial v_i}{\partial x_i}\frac{\partial f_i}{\partial v_i} \\ &= \frac{\partial f_i}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial f_i}{\partial v_i} \end{aligned} \tag{6}$$

[redacted] [redacted] $v_i \equiv \frac{x_i - x_{i-1}}{\varepsilon}$ [redacted]
[redacted] $\frac{\partial f}{\partial x_i}$ [redacted] f_{i+1} [redacted]

$$\begin{aligned} \frac{\partial}{\partial x_i} f_{i+1} &= \frac{\partial}{\partial x_i} f\left(x_{i+1}, \frac{x_{i+1} - x_i}{\varepsilon}\right) \\ &= \frac{\partial v_{i+1}}{\partial x_i}\frac{\partial f_{i+1}}{\partial v_{i+1}} \\ &= -\frac{1}{\varepsilon}\frac{\partial f_{i+1}}{\partial v_{i+1}} \end{aligned} \tag{7}$$

[redacted]

$$\frac{\partial f_i}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial f_i}{\partial v_i} - \frac{1}{\varepsilon}\frac{\partial f_{i+1}}{\partial v_{i+1}}. \tag{8}$$

[redacted]

[redacted] $\frac{f_{i+1} - f_i}{\varepsilon}$ [redacted] $\varepsilon \to 0$ [redacted] [redacted] v_i [redacted]
[redacted]

$$\frac{\partial f_i}{\partial x_i} - \frac{d}{dt}\frac{\partial f_i}{\partial \dot{x}_i} = 0 \tag{9}$$

[redacted] [redacted] i [redacted] t [redacted]

$$\begin{aligned} x &= X\cos(\omega t) + Y\sin(\omega t) \\ y &= X\sin(\omega t) - Y\cos(\omega t). \end{aligned} \tag{10}$$

[redacted] [redacted]
[redacted] [redacted]
[redacted] [redacted]

[redacted]

$$\mathcal{L} = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) \tag{11}$$

[redacted]

[redacted] (2) [redacted]

$$\mathcal{L} = \frac{m}{2}\left(\dot{X}^2 + \dot{Y}^2\right) + \frac{\omega^2 m}{2}(X^2 + Y^2) + \frac{\omega m}{2}\left(X\dot{Y} - Y\dot{X}\right) \tag{12}$$

[redacted]

[redacted] [redacted] $r = X^2 + Y^2$ [redacted] r [redacted] [redacted]
[redacted]

[redacted] X [redacted]

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial X} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{X}} &= \omega^2 m X + \frac{\omega m}{2}\dot{Y} - m\ddot{X} + \frac{\omega m}{2}\dot{Y} = 0 \\ \therefore m\ddot{X} &= \omega^2 m X + \omega m \dot{Y} \end{aligned} \tag{13}$$

[redacted] $ma = F$ [redacted] X [redacted] [redacted] Y [redacted]
$\dot{X}$ [redacted] [redacted] [redacted]
[redacted] [redacted]

1.3. [redacted]

[redacted] [redacted] r, θ [redacted] [redacted]
[redacted]

$$\begin{aligned} x &= r\cos(\theta) \\ y &= r\sin(\theta). \end{aligned} \tag{14}$$

[redacted]

$$\begin{aligned} \mathcal{L} &= \frac{m}{2}(\dot{x}^2 + \dot{y}^2) \\ &= \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) \end{aligned} \tag{15}$$

[redacted]

[redacted] r [redacted]

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial r} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{r}} &= mr\dot{\theta}^2 - \frac{d}{dt}m\dot{r} \\ &= mr\dot{\theta}^2 - m\ddot{r} = 0 \\ \therefore mr\dot{\theta}^2 &= m\ddot{r}. \end{aligned} \tag{16}$$

[redacted] $F = ma$ [redacted] θ^2 [redacted] [redacted] r [redacted]2[redacted]

[redacted] θ [redacted]

$$\frac{\partial \mathcal{L}}{\partial \theta} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = -mr^2\dot{\theta} = 0. \tag{17}$$

[illegible]

[illegible]

1.4. [illegible]

[illegible]

[illegible]

$$\delta f(\vec{q}) = \sum_i \frac{\partial f}{\partial q_i}\delta q_i \tag{18}$$

[illegible] $f(\dot{q})$ [illegible] $q_{i(i=1,2,\ldots,n)}$ [illegible]

[illegible]

[illegible] q_i [illegible] δq_i [illegible] $f_{i(\dot{q})}$ [illegible]

$$\delta q_i = f_i(\vec{q})\delta \tag{19}$$

[illegible] δ [illegible] δ [illegible]

[illegible] q_i [illegible] $\dot{q}_i$ [illegible]

$$\delta \mathcal{L}\left(\vec{q}, \dot{\vec{q}}\right) = \sum_i \left(\frac{\partial \mathcal{L}}{\partial q_i}\delta q_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i\right). \tag{20}$$

[illegible]

[illegible] Equation 1 [illegible]

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \equiv p_i. \tag{21}$$

[illegible]

[illegible]

$$\frac{\partial \mathcal{L}}{\partial q_i} = \dot{p}_i \tag{22}$$

[illegible]

[illegible] Equation 20 [illegible]

$$\delta \mathcal{L}\left(\vec{q}, \dot{\vec{q}}\right) = \sum_i \left(\dot{p}_i \delta q_i + p_i \delta \dot{q}_i\right) \tag{23}$$

[illegible]

$$\delta \mathcal{L}\left(\vec{q}, \dot{\vec{q}}\right) = \frac{d}{dt}\sum_i p_i \delta q_i \tag{24}$$

[illegible]

[illegible] δq_i [illegible] Equation 19 [illegible]

$$\delta \mathcal{L}\left(\vec{q}, \dot{\vec{q}}\right) = \frac{d}{dt} \sum_i p_i \delta q_i \tag{25}$$

[illegible]

δ [illegible]

$$\frac{d}{dt} \sum_i p_i f_i(\vec{q}) = 0 \tag{26}$$

[illegible]

[illegible]

$$Q \equiv \sum_i p_i f_i(\vec{q}) \tag{27}$$

[illegible]

[illegible]

[illegible]

2. [illegible]

[illegible]

[illegible] $q_i, \dot{q}_i$ [illegible] [illegible] $\mathcal{L}(\dot{q}, \ddot{q})$ [illegible]

[illegible]

$$\frac{d\mathcal{L}\left(\vec{q}, \dot{\vec{q}}\right)}{dt} = \sum_i \left\{ \frac{\partial \mathcal{L}}{\partial q_i} \dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \ddot{q}_i \right\} \tag{28}$$

[illegible] Equation 21, Equation 22 [illegible]

$$\frac{d\mathcal{L}}{dt} = \sum_i \left(p_i \dot{q} + \dot{p}_i \ddot{q}_i \right). \tag{29}$$

[illegible]

$$\frac{d\mathcal{L}}{dt} = \sum_i \frac{d}{dt} p_i \dot{q}_i \tag{30}$$

[illegible]

$$\frac{d}{dt} \left\{ \mathcal{L} - \sum_i p_i \dot{q}_i \right\} = 0 \tag{31}$$

[illegible]

[illegible]

$$\mathcal{H} \equiv \sum_i p_i \dot{q}_i - \mathcal{L} \tag{32}$$

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3. Leonard Susskind, Classical Mechanics | Lecture 5
4. Wikipedia's Noether's theorem article